

Virtual Escape Room Application

Software Requirements Specification



CSCE 247: Software Engineering

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Tech Titans

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1. Introduction

Purpose

When one hears the word “escape room”, they might think of a situation where a group of people need to come up with a plan to escape, which is correct. An escape room is a live-action game in which a group of 2 or more people are placed in an environment and must escape within a given time. The disadvantage is that most escape rooms have restrictions like age, group size, price, and more. According to The Final Door in Columbia, the price for a group of two is between \$42.99 and \$55.99, while groups of three or more are \$36.99. While these escape rooms are highly rated, it comes with obstacles like children (usually under 16) not being old enough to participate, cost, and even location. They also need to be booked far in advance. This is where our group, Tech Titans, comes in. People of all ages should be able to experience the exciting escape rooms have to offer. With our virtual escape room, things like costs, booking ahead, and transportation are no longer barriers to having the experience. Being able to learn valuable skills like critical thinking, problem solving, and teamwork that are important for future careers should be accessible to everyone.

Source: <https://www.thefinaldoor.com/>

Scope

- The personas of potential users and stakeholders invested into this project.
- Any constraints that have been applied to this project.
- A description of the website/app along with its business use cases.
- Shows both the functional and nonfunctional requirements of the app.
- A competitive analysis to outline the purpose of the project

2. Stakeholders

- Mostly people who have some experience in problem solving and puzzles.
 - It can be due to work or hobbies.
- People who are looking for fun activities and things to do in their free time.
 - This can be alone or with friends and coworkers.
- People who want to be challenged or pushed to solve things.

Personas

MICHEAL STONE

PROFILE

Gender : Male
Age : 28
Education : Bachelor's degree
Occupation : Data Analyst
Character : Leader



BIOGRAPHY

Michael is a Data Analyst who works for a investment bank. He enjoys spending time with his dogs and going to work. He likes to try out different restaurants and watch old films as well

MOTIVATIONS

Seeks solutions to things with a problem solving mindset, Michael wants his coworkers to feel connected and foster friendships. He craves thrill and excitement in his life, something besides his work.

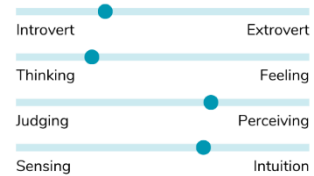
GOALS

Wants collaboration and communication with coworkers
Wants to engage in team building activities
Wants to find more things to do outside of work
Organize his thoughts and skills

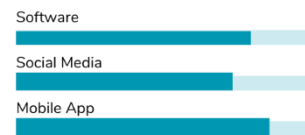
FRUSTRATIONS

Struggles to find activities that suit him
Has a hard time keeping his mind off work
Dislikes non-intuitive interfaces
Most things are not thrilling enough in his life

PERSONALITY



TECHNOLOGY



BRANDS



AVA REYES

PROFILE

Gender : Female
Age : 24
Education : Master's in Behavioral Psychology
Occupation : Puzzle designer & cognitive researcher
Character : Visionary



BIOGRAPHY

Ava creates immersive puzzle experiences to explore how people think. She's a jazz pianist and urban explorer who draws inspiration from forgotten places and stories.

MOTIVATIONS

She's driven by the belief that every puzzle should feel like a personal journey one that lingers long after the final lock clicks open. Turning gameplay into transformation

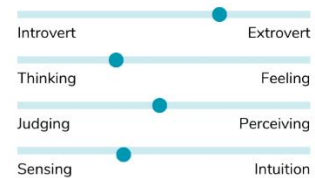
GOALS

Study how people think under pressure
Craft puzzles that invoke emotion for audience with insight.
Maintain balance between creativity and self care
Travel to deepen her understanding of human thinking

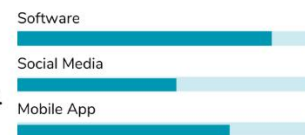
FRUSTRATIONS

Rushing through puzzles without reflection
Difficulty letting go of creative control
Repetitive puzzle format with no depth
Most things are not thrilling enough in his life

PERSONALITY



TECHNOLOGY



BRANDS



MELISSA REYNOLDS

PROFILE

Gender	Female
Age	20
Education	Criminology & Criminal Justice Major
Occupation	College Student
Character	Enigma Chaser



BIOGRAPHY

Melissa has always enjoyed solving puzzles and mysteries. She is naturally detail-oriented and likes to view situations from multiple perspectives. She creates new ideas by combining elements that others wouldn't.

MOTIVATIONS

She believes that every problem or situation has a solution, regardless of how tough or lengthy the process may be. Her favourite phrase is "If every action has an equal and opposite reaction, then every problem has an equal and opposite solution."

GOALS

- Discovering strategies for solving similar situations.
- Plays Sudoku for 1 hour daily to sharpen her critical thinking.
- See the Northern lights in Alaska.
- Visit the 3 pyramids of Egypt.

FRUSTRATIONS

- Dealing with decimals (e.g. \$17.68, make it \$20).
- Dealing with puzzles with a random pattern.
- Having items out of place (e.g. row of cups in different sizes)
- Things done halfway (e.g. eating without washing up)

PERSONALITY

Introvert	Extrovert
Thinking	Feeling
Judging	Perceiving
Sensing	Intuition

TECHNOLOGY

Software	
Social Media	
Mobile App	

BRANDS




3. Constraints

Time Constraints:

- Project must be finished within a span of weeks.
- Must collaborate with team members, which may limit the time.

Language and Coding Constraints

- Must be done in Java, can include other languages, however mainly Java.
- Must include Javadoc and easy to read code.
- Must work on different devices, must be accessible.

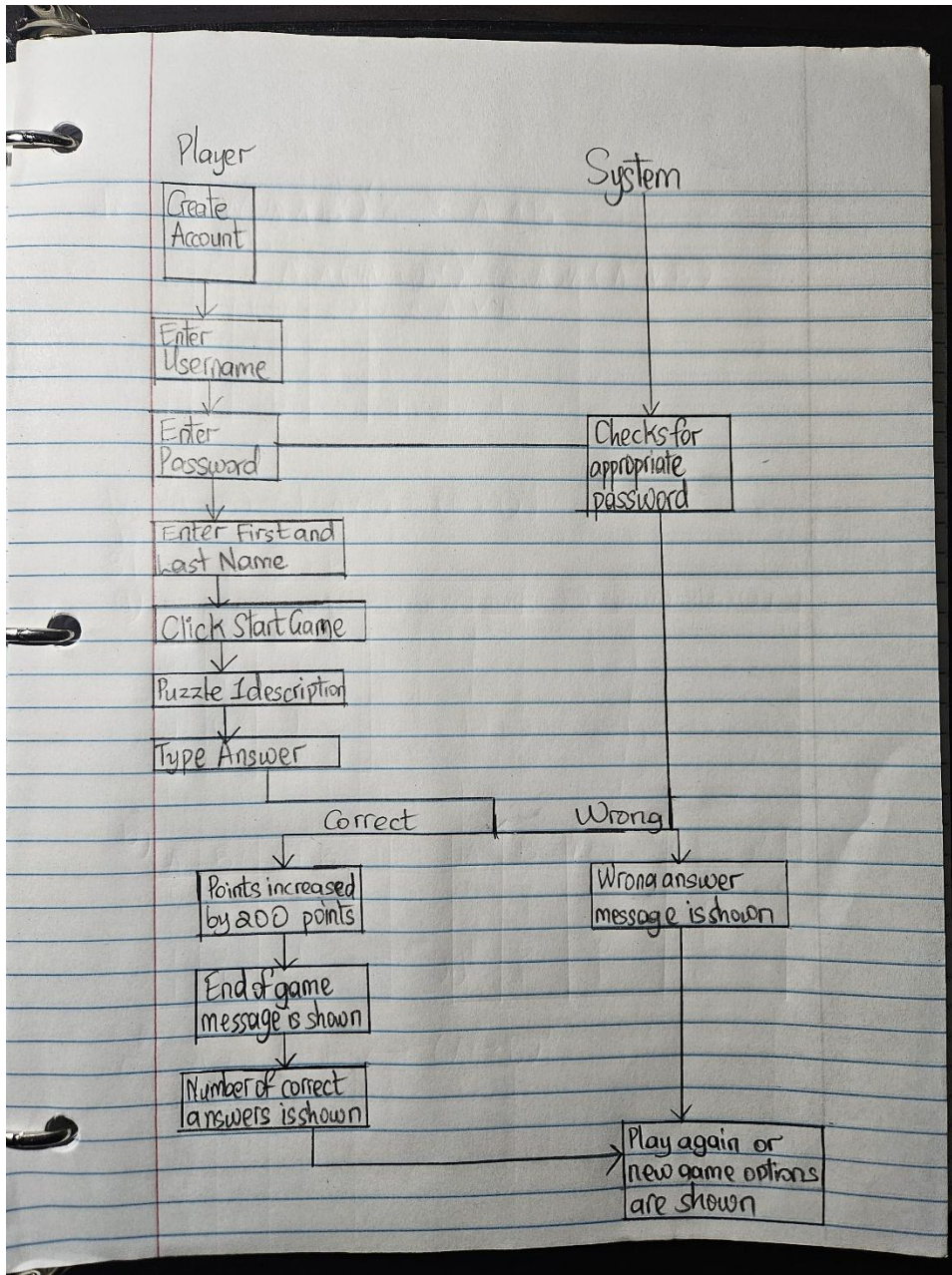
Financial Constraints

- Development has a budget of \$0.

4. Overall Description

The app must be easily accessible to all people. Players will have to navigate through different views within the application. The website may collect some data for fast completion times and for bug reports. The Escape room experience will be created using Java and JavaFX, for primarily PCs. It will be an immersive experience where the user will have to go through challenges where they solve problems and obstacles to escape.

5. Business Use Cases



6. Functional Requirements

[Spreadsheet for requirements](#)

7. Non-Functional Requirements

Security Requirements

- All user data shall be encrypted to ensure user privacy and security.
- User information shall be stored.

Performance Requirements

- The user shall be able to use the application on most laptops.
- Users do not need Wi-Fi for application use.

Usability Requirements

- The user shall be able to navigate application with ease.
- The system shall be intuitive to the user.

Legal Requirements

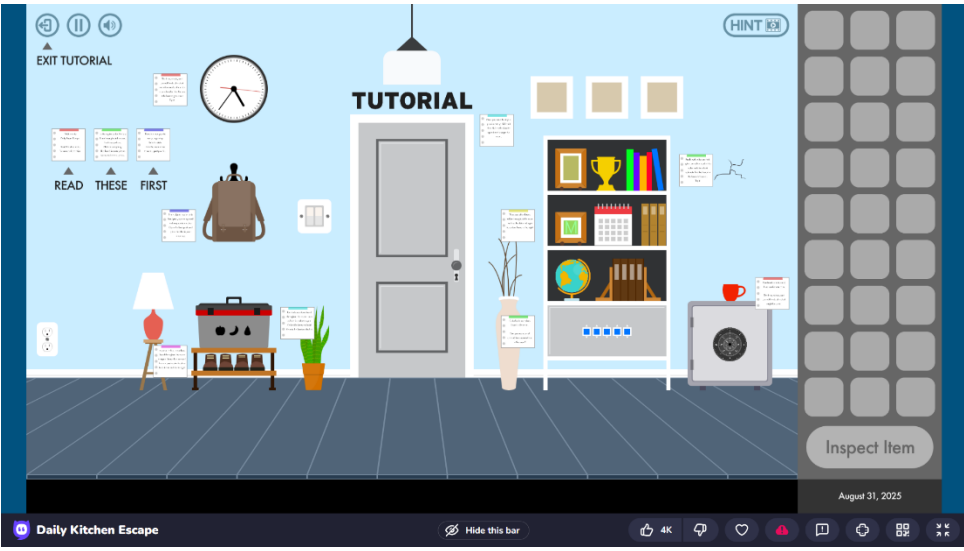
- The system shall comply with existing laws and regulations
 - State or local laws

9. Competitive Analysis

Implications

Through our competitive analysis we have drawn conclusions on what we would like to see in our virtual escape room. There should be a thematic element or storyline that matches the challenges and puzzles. Furthermore, we would like to incorporate effects to confirm our thematic elements. Clues and challenges should have clear instructions and be well thought out. Instructions in general should be easy to read and accessible for those with impairments. No sign in or money required to play.

	Game 1
Picture	 Your darling Juliet, is with her family. You want to talk with her parents and try to convince them to accept that Juliet and you live your love without problems. Juliet parents won't make this easy and they want that you demonstrate you are enough smart to deserve her intelligent daughter. Will you be able to do it http://romeojuliet.heniak.com/
Strengths	It is heavily dependent on the storyline and feels very thematic, lets you take over characters roles, such as Romeo. The challenges are fun but not too challenging. Allows you to hear clues through audio. Allow for spoken instructions which are inclusive and accessible for disabilities. It is easily accessible and free for use, and does not require email or signing in. It is efficient in terms of speed and does not lag. The visuals and instructions are clear and big, which makes them easy to read.
Weaknesses	Does not have a back button to go back to previous challenges or instructions. Most of the challenges were at the same level of difficulty, there was no progression. In addition, some of the hints were not very helpful and were redundant. The input format was unclear and confusing at times.

Audience/Focus	The audience could include Shakespeare or literature fans, people who enjoy solving puzzles, educators and perhaps gamers. Most likely not kids under 12, as some of the challenges may be confusing for them.
	Game 2
Picture	 https://www.crazygames.com/game/daily-kitchen-escape
Strength	It has two categories, a tutorial and the main game. The tutorial serves as a warmup for those who may have never played in escape rooms or escape rooms on its platform. It allows the player to click every object in the room (s) which is like an in-person escape room. It had multiple types of clues like 3, and 4-digit number combinations, and color combinations as well which allows the user to view things from different perspectives.

Weaknesses	Some of its clues were difficult to comprehend. For example, it had a binary code sheet as a clue, but it was difficult to apply the binary in the game itself. Clues like the binary code sheet might be confusing for kids 12 and under if they have not learnt about binary tables yet.
Audience/Focus	It does not have a specific target. It is intuitive and has a simple/childlike design. The objects are easy to spots, and the colors make it more appealing to the mind, making it suitable for all ages.

	Game 3
Picture	 https://store.steampowered.com/app/1196590/Resident_Evil_Village/
Strengths	• This game has exceptional graphics and storyline • They utilize horror elements to allow for less complex puzzles that become harder to solve without a generic timer • Not traditionally an Escape Room, but utilizes the same concepts very well

Weaknesses	• Costs money • Not everyone is going to play a horror game or one as long as this one is.
Audience/Focus	• This one is to general audiences rated M for 17+, but the game is for anyone who enjoys horror or the Resident Evil Franchise